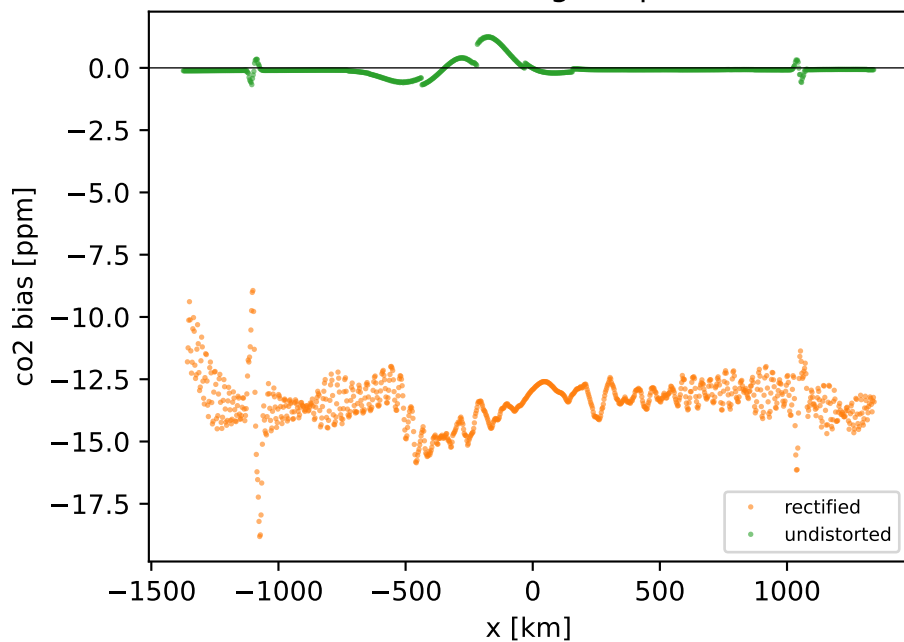
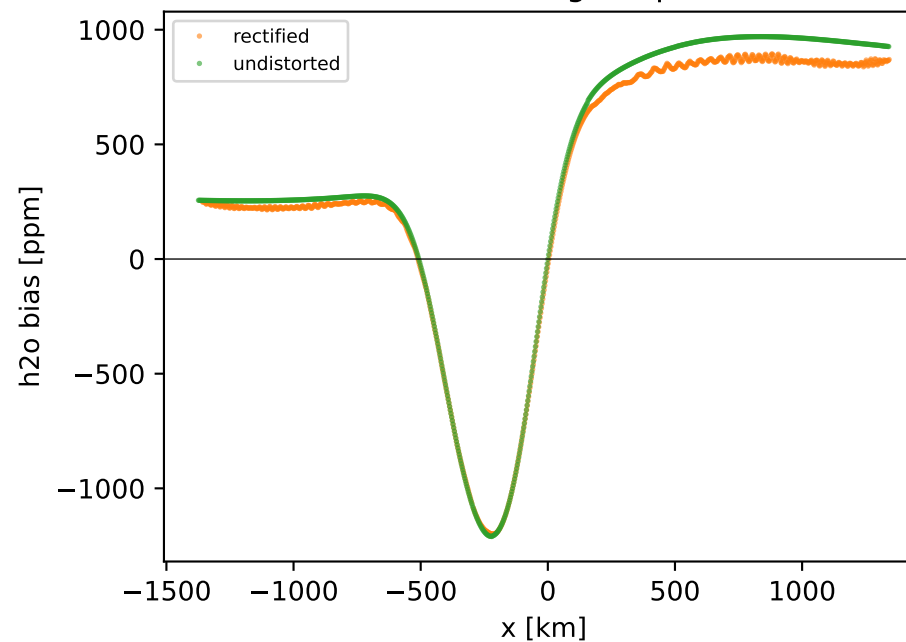


Joint FPA0+FPA1+FPA2 -- realistic, no noise (rectified/undistorted)

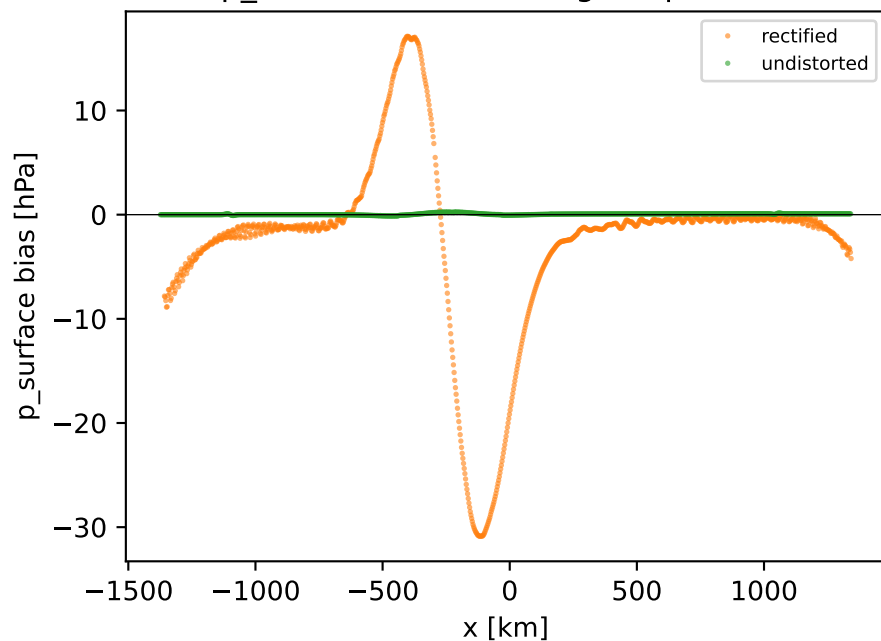
co2 bias vs along-slit position



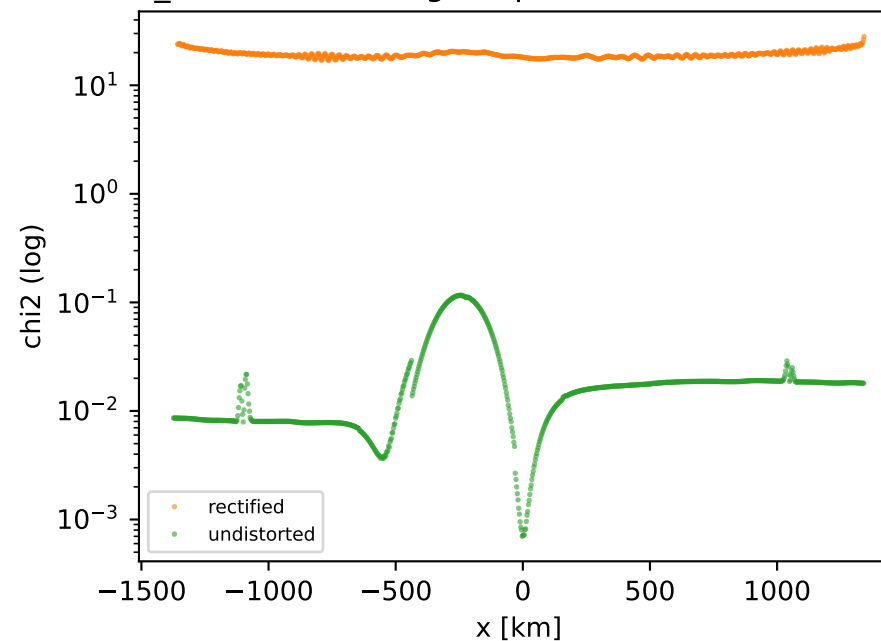
h2o bias vs along-slit position



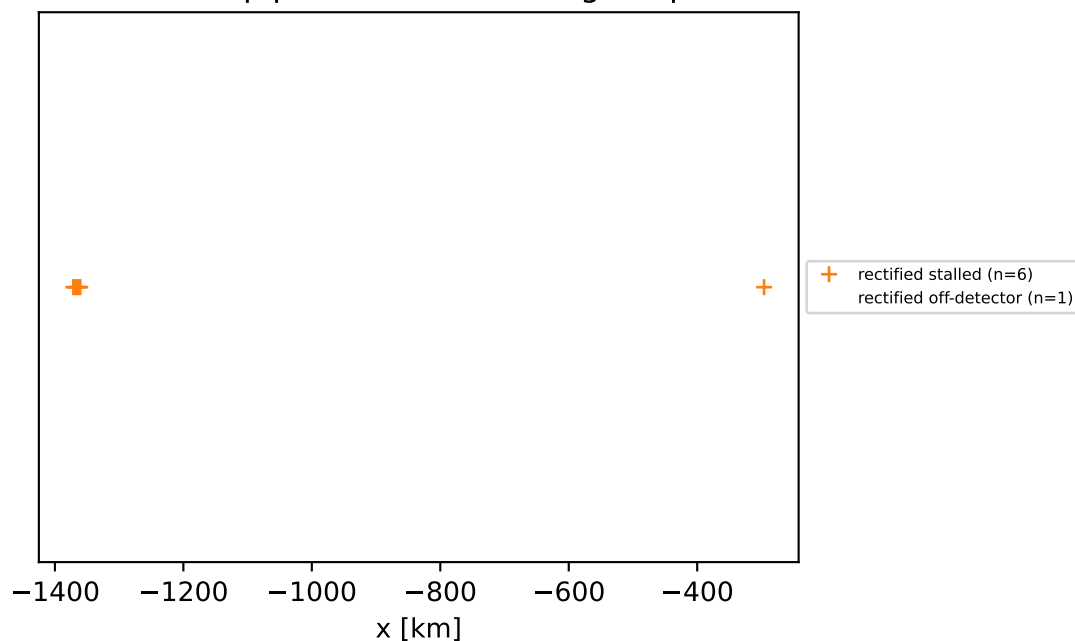
p_surface bias vs along-slit position



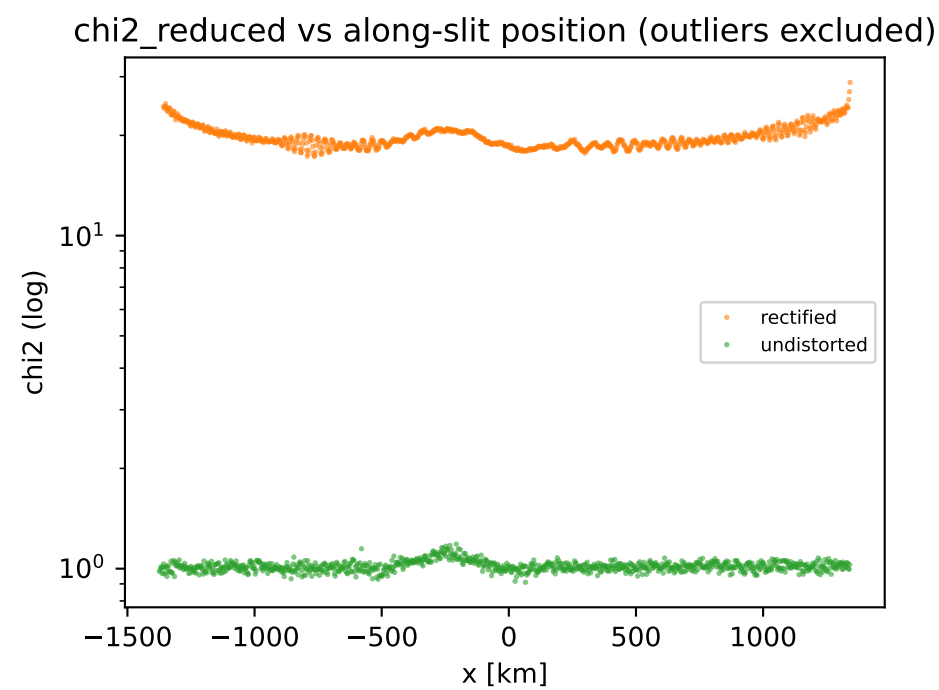
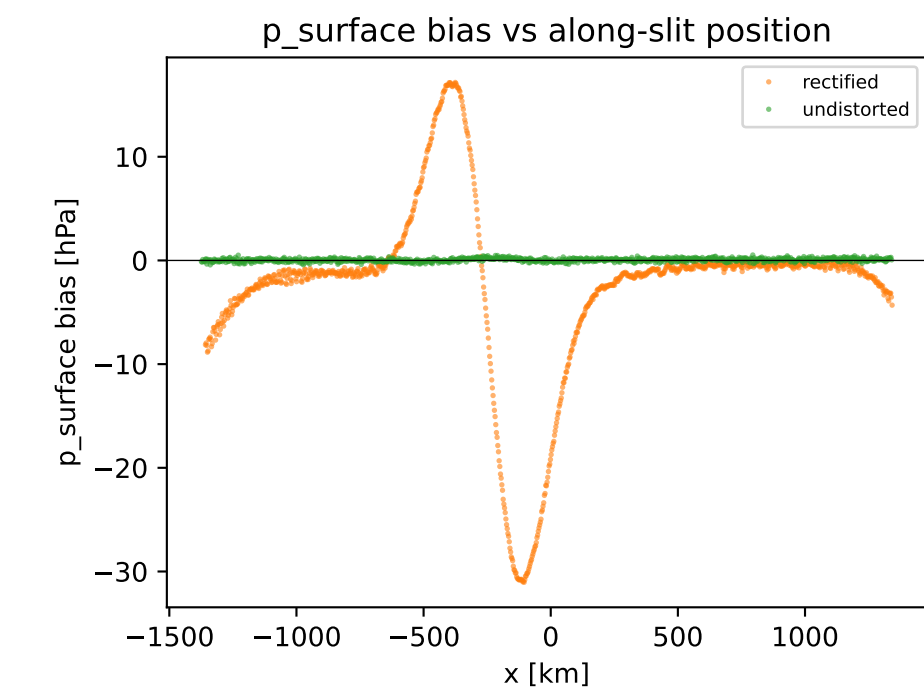
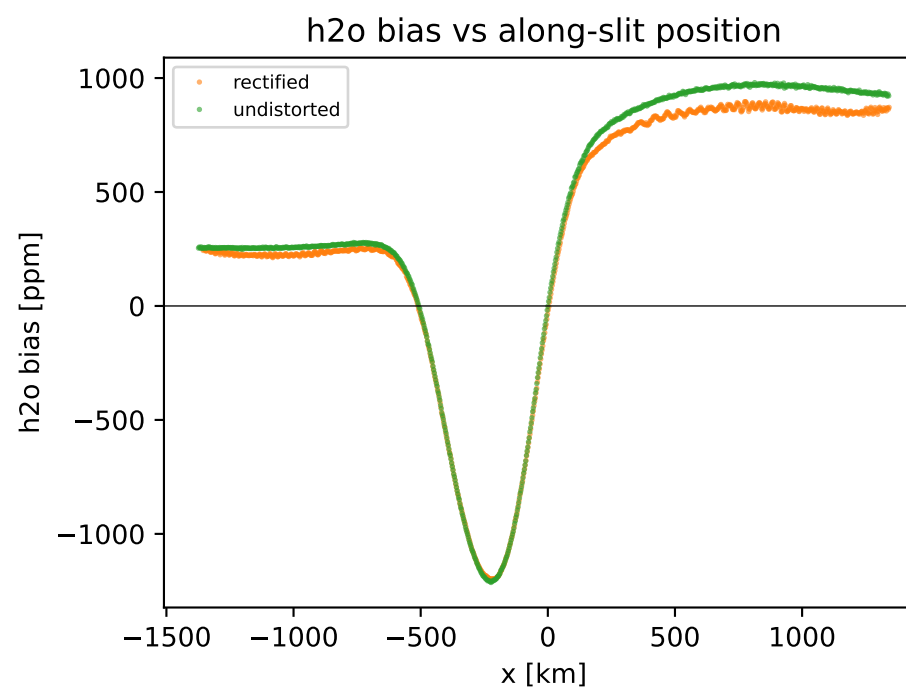
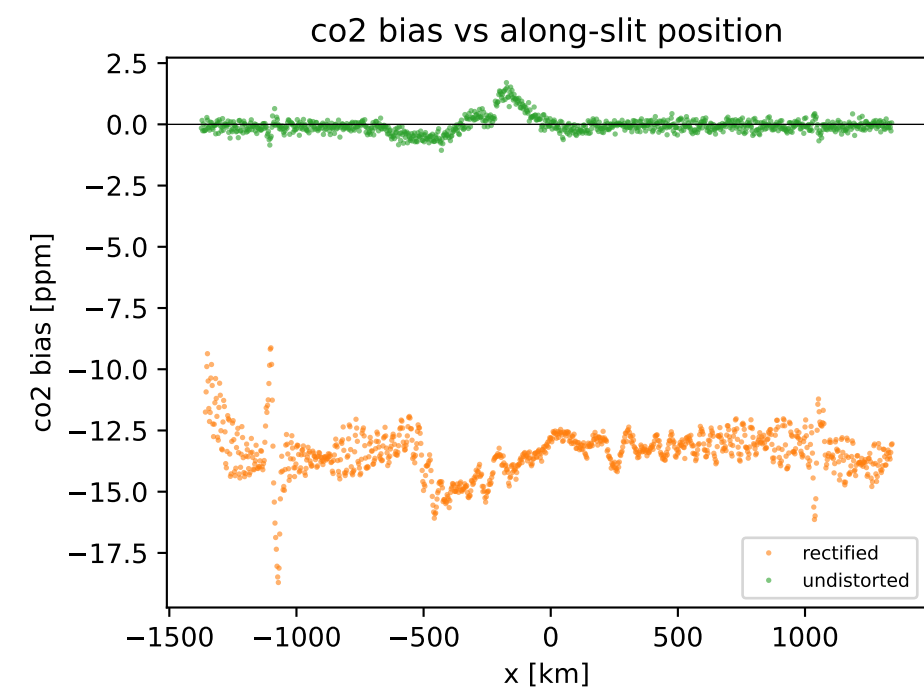
chi2_reduced vs along-slit position (outliers excluded)



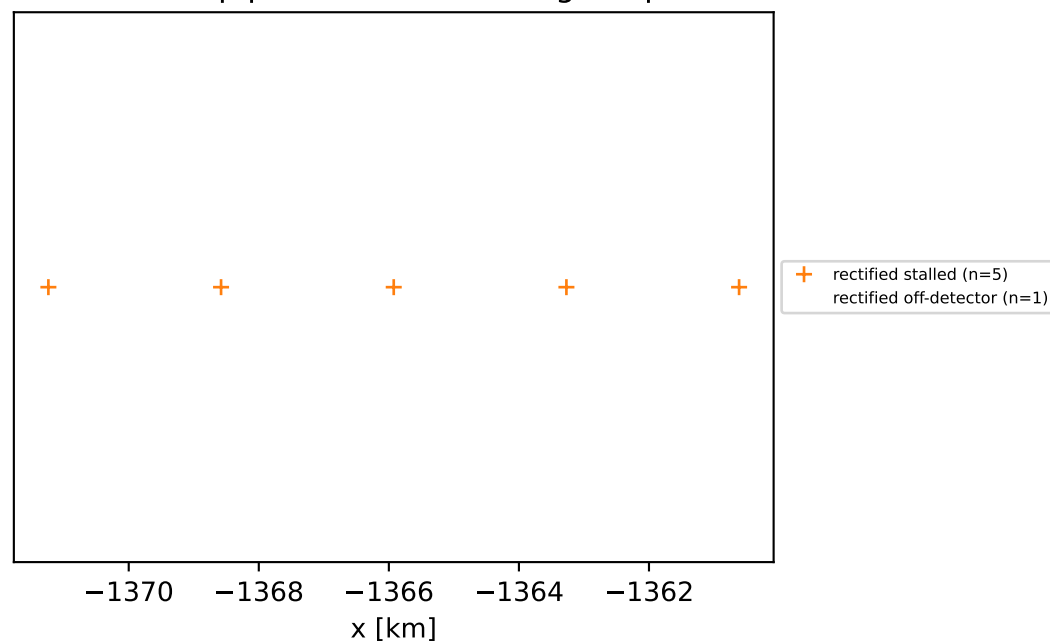
Where each pipeline fails, vs along-slit position



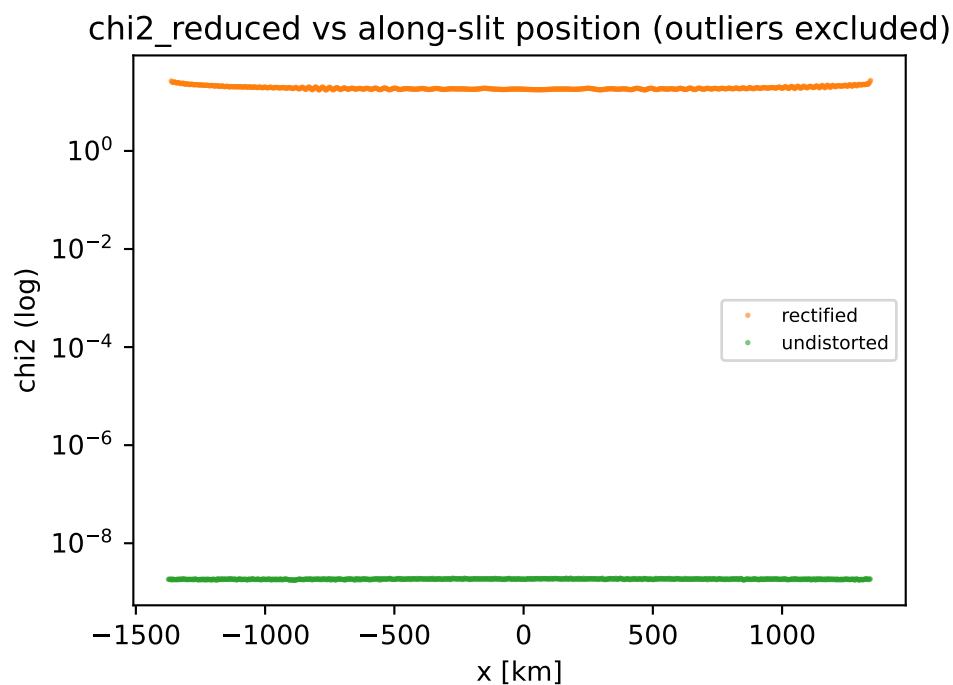
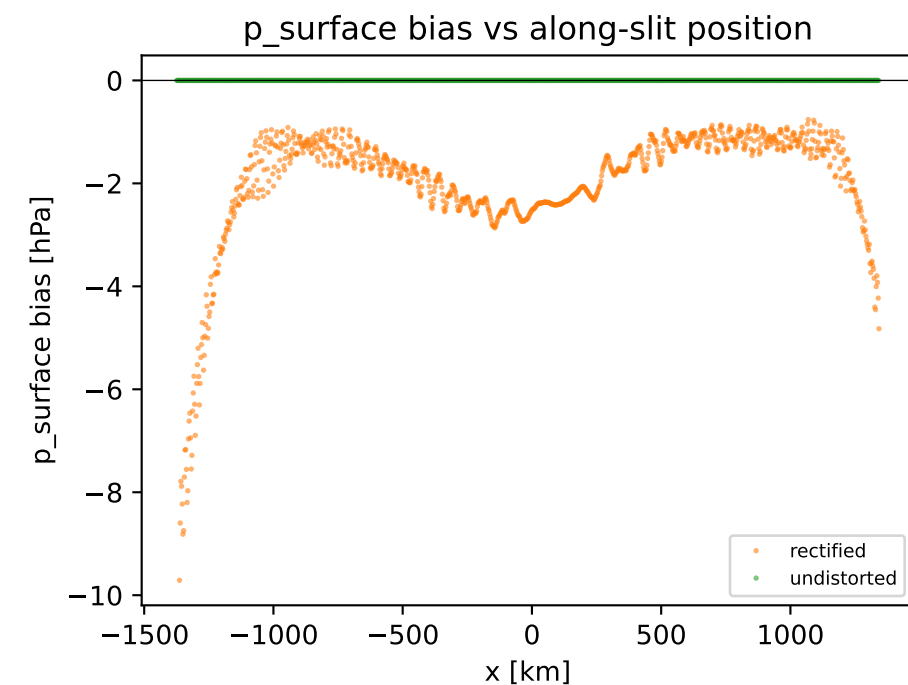
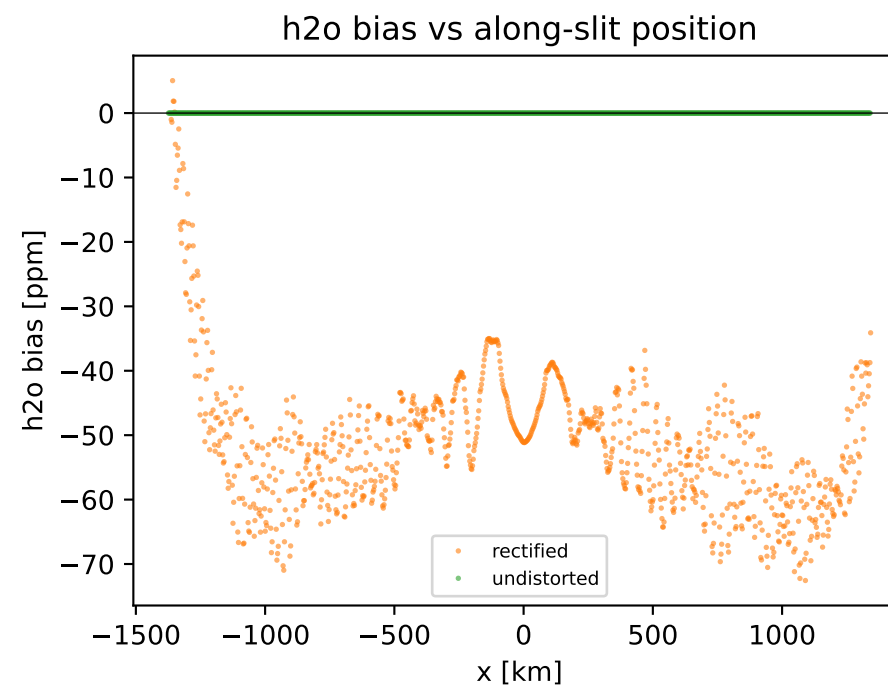
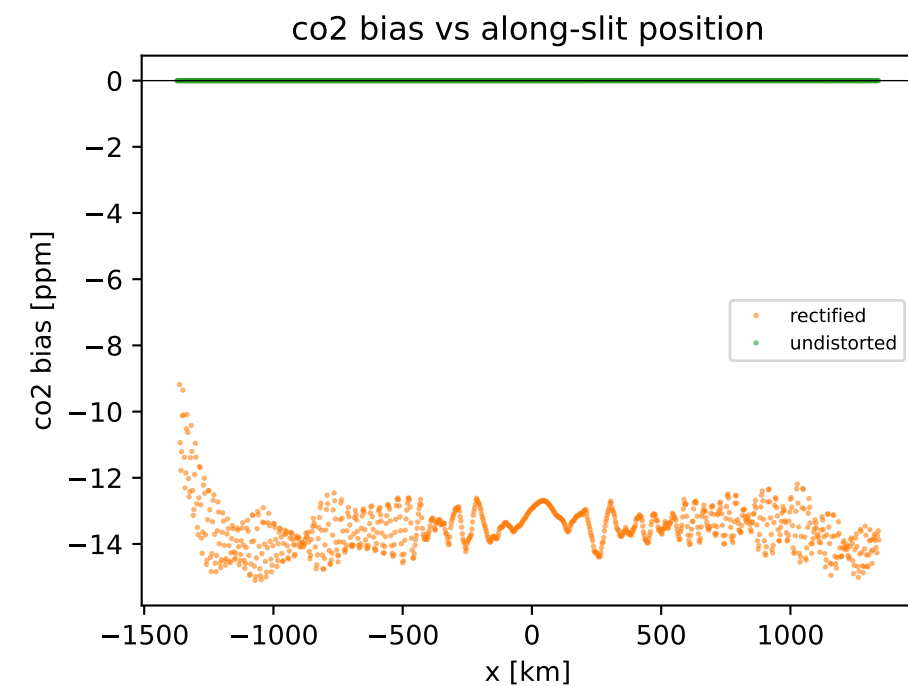
Joint FPA0+FPA1+FPA2 -- realistic, noise (rectified/undistorted)



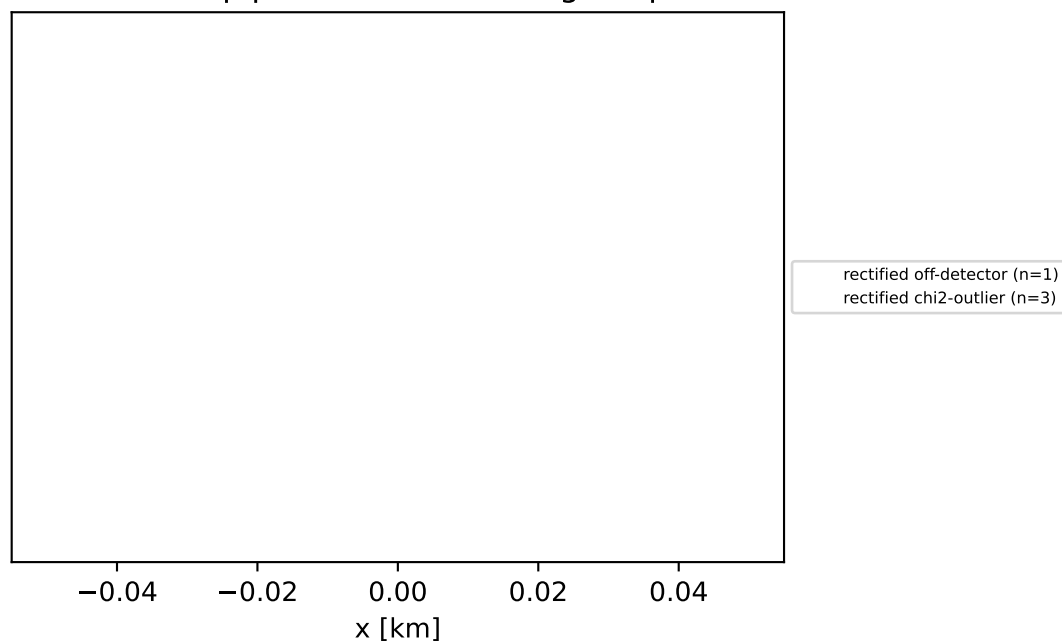
Where each pipeline fails, vs along-slit position



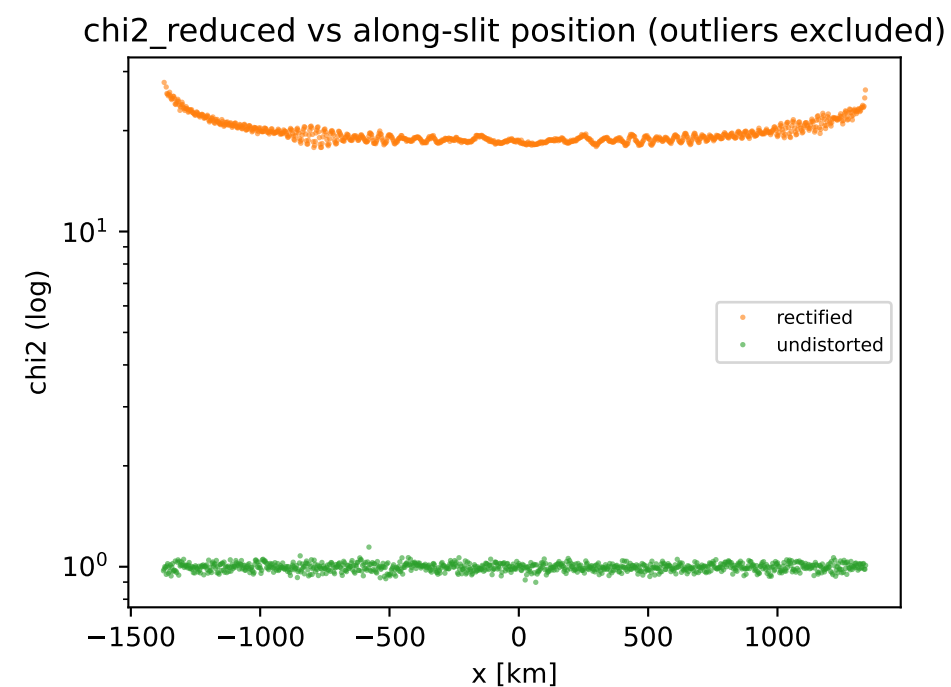
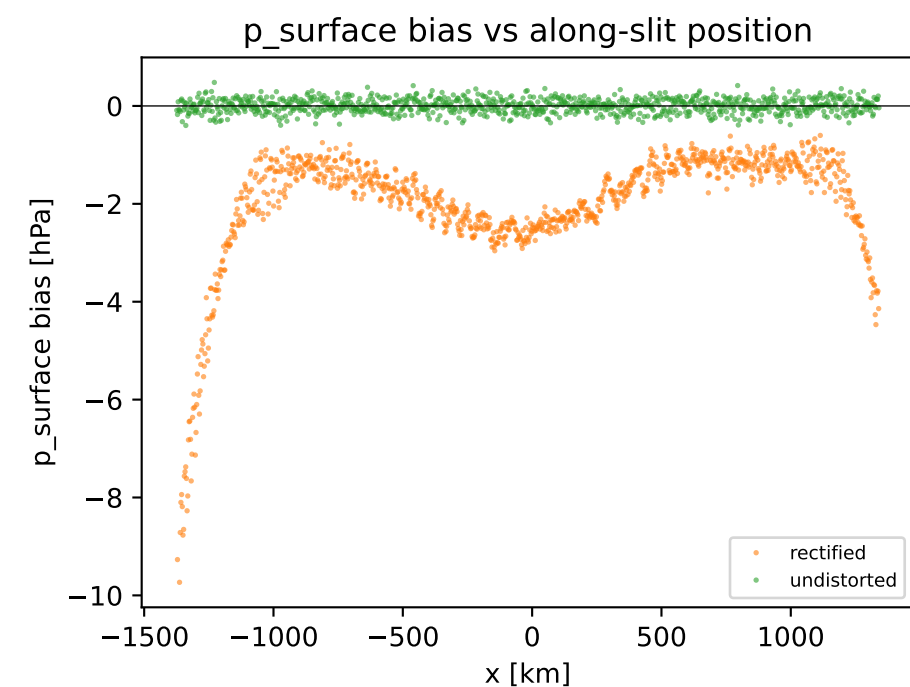
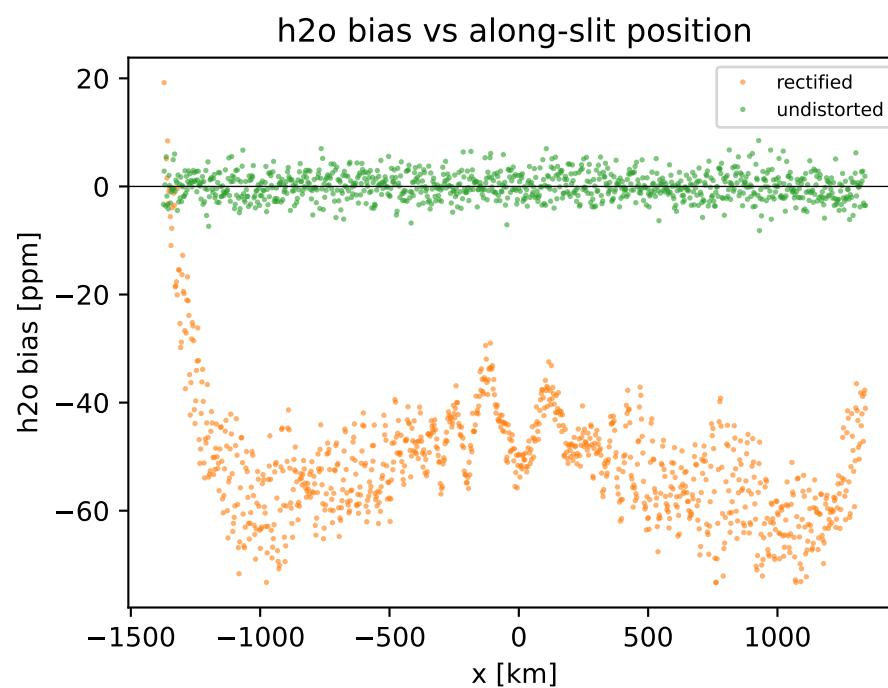
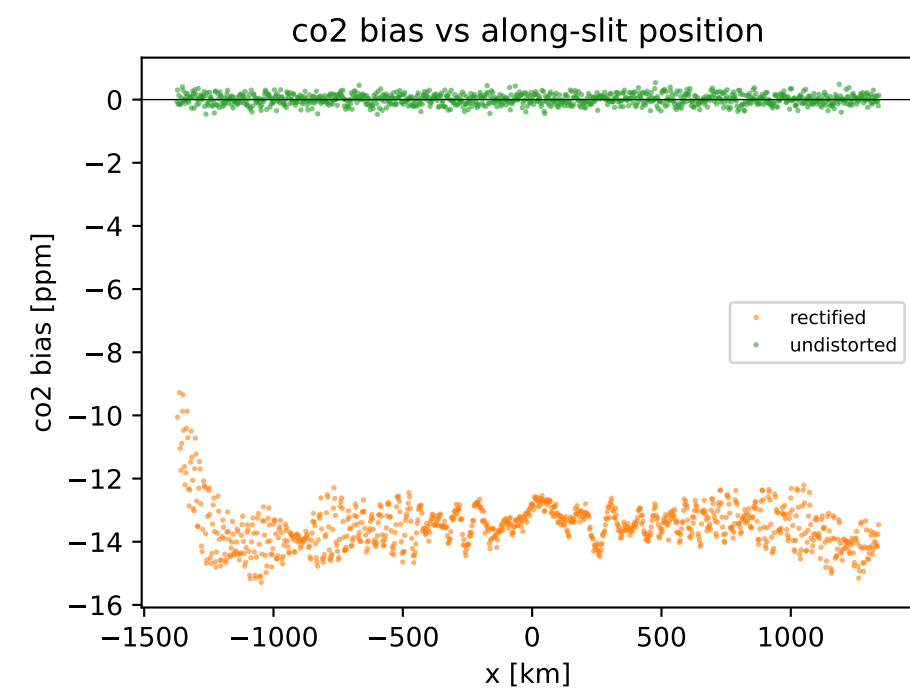
Joint FPA0+FPA1+FPA2 -- uniform, no noise (rectified/undistorted)



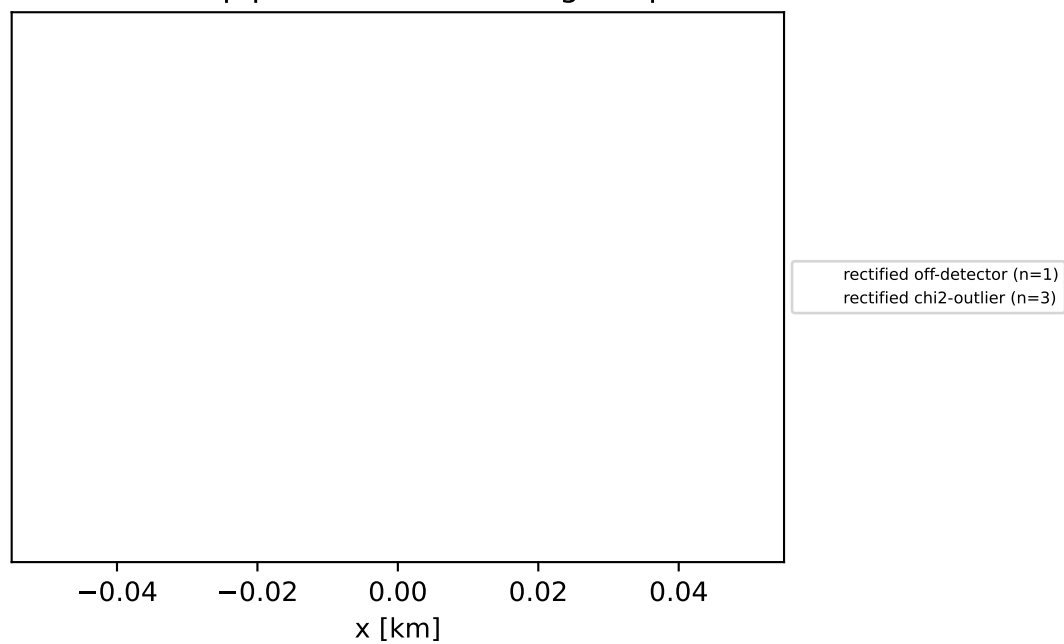
Where each pipeline fails, vs along-slit position



Joint FPA0+FPA1+FPA2 -- uniform, noise (rectified/undistorted)

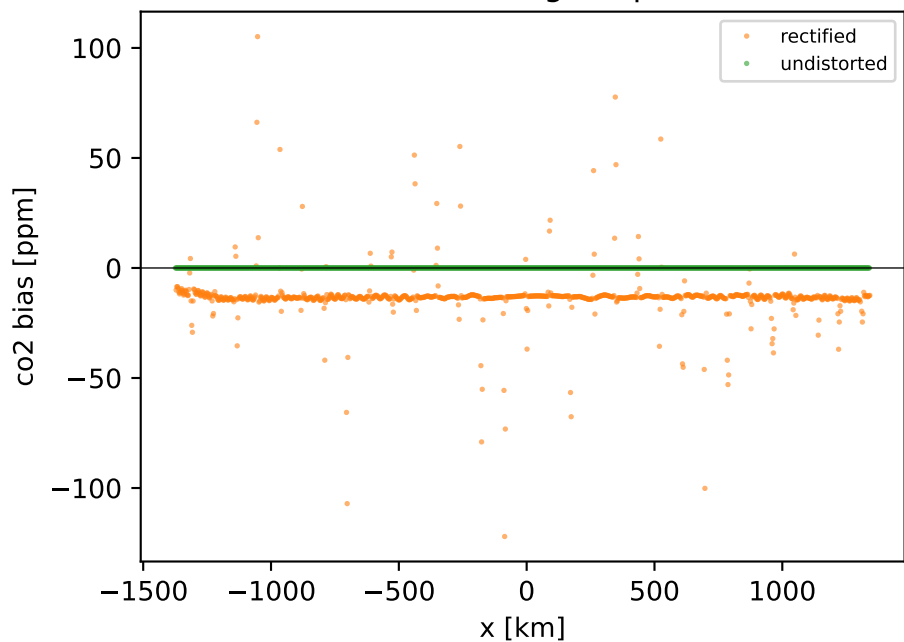


Where each pipeline fails, vs along-slit position

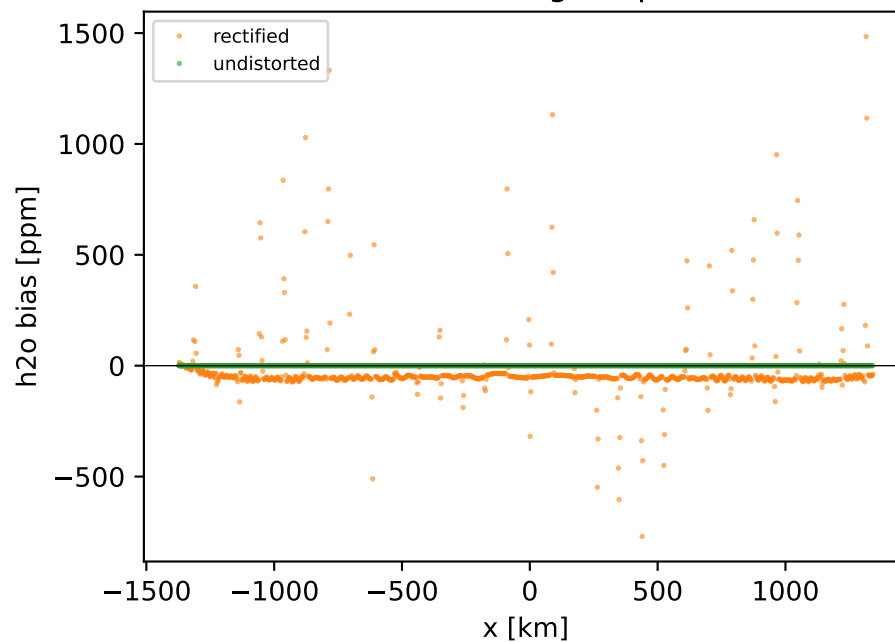


Joint FPA0+FPA1+FPA2 -- barcode, no noise (rectified/undistorted)

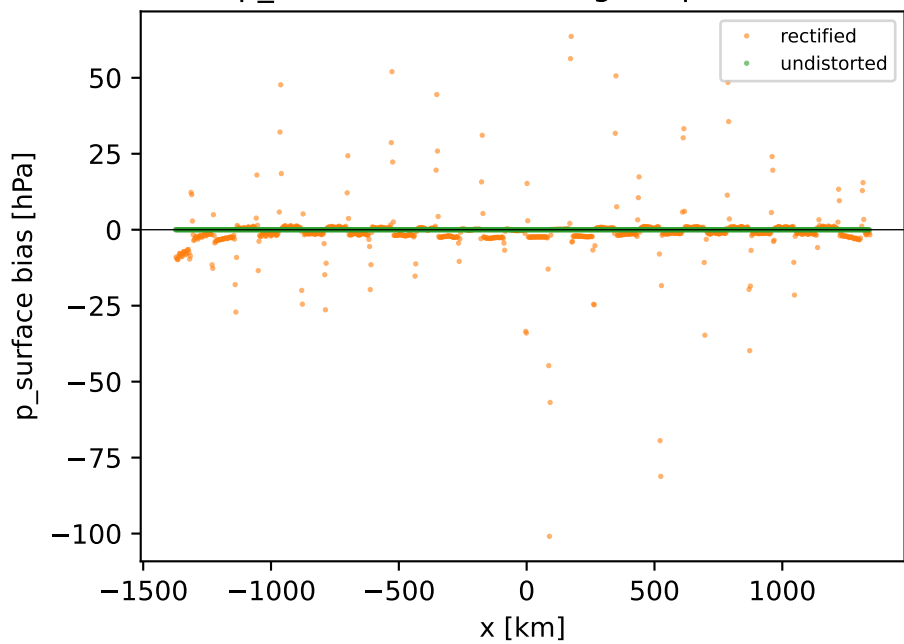
co2 bias vs along-slit position



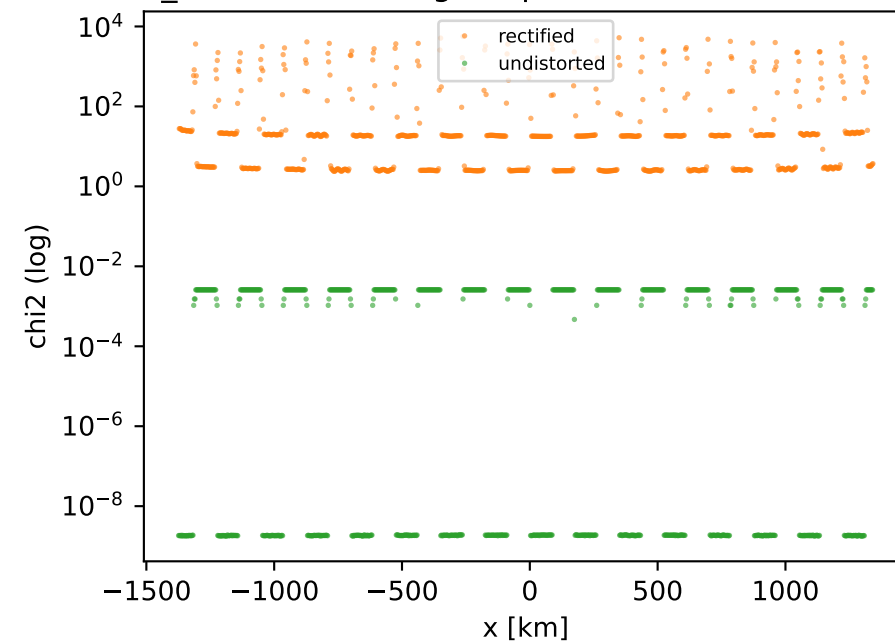
h2o bias vs along-slit position



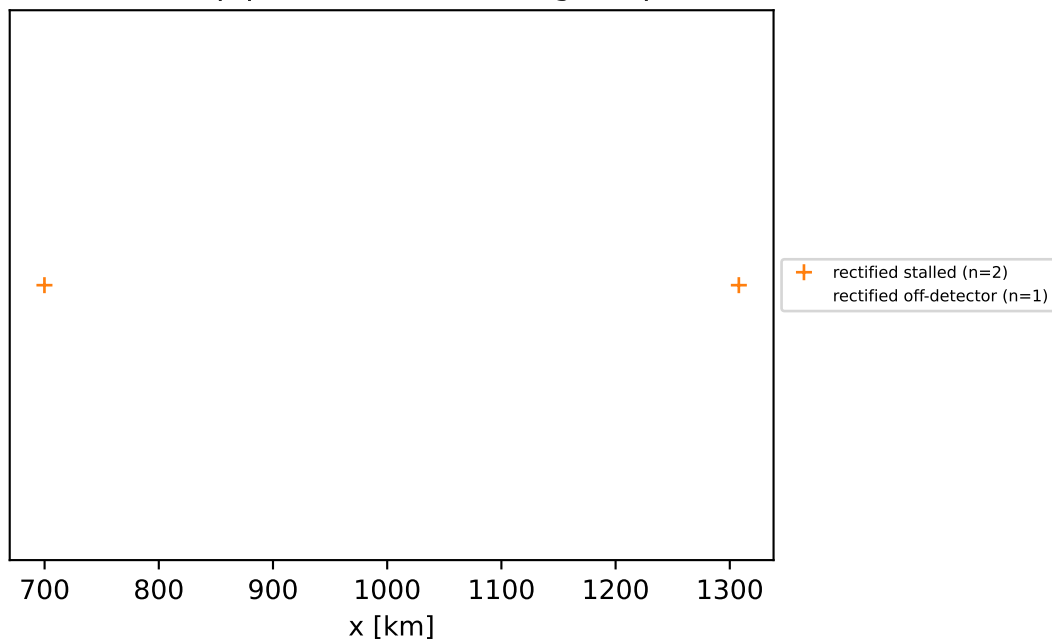
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

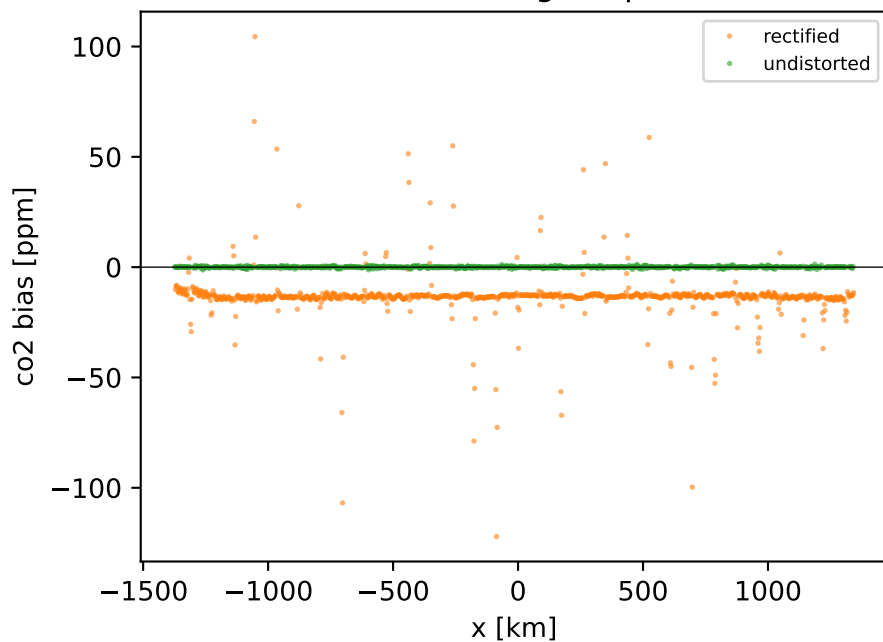


Where each pipeline fails, vs along-slit position

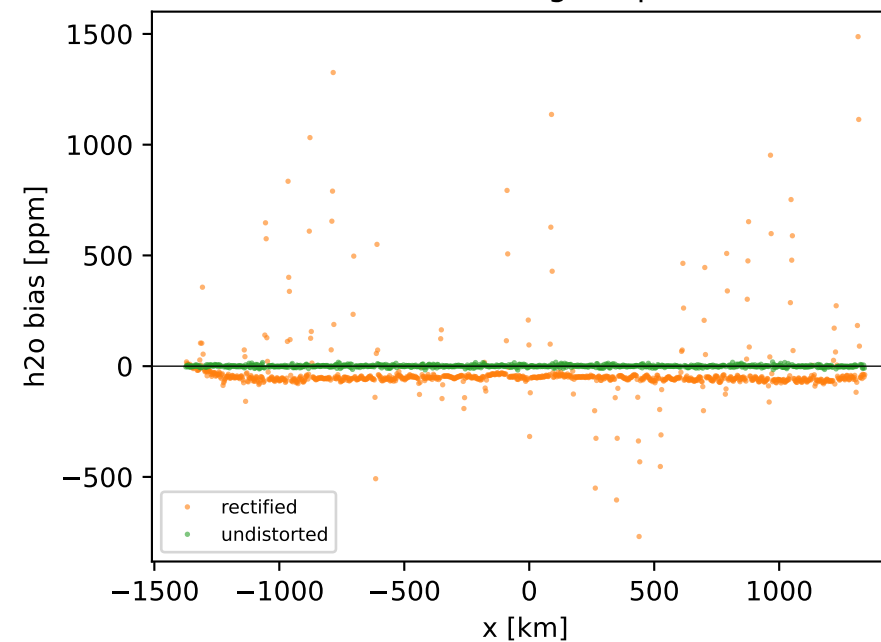


Joint FPA0+FPA1+FPA2 -- barcode, noise (rectified/undistorted)

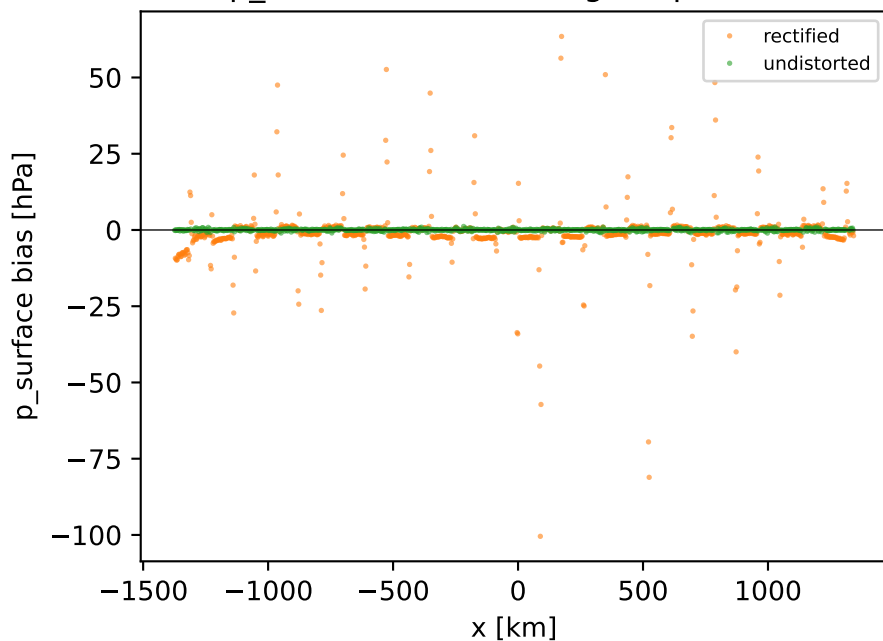
co2 bias vs along-slit position



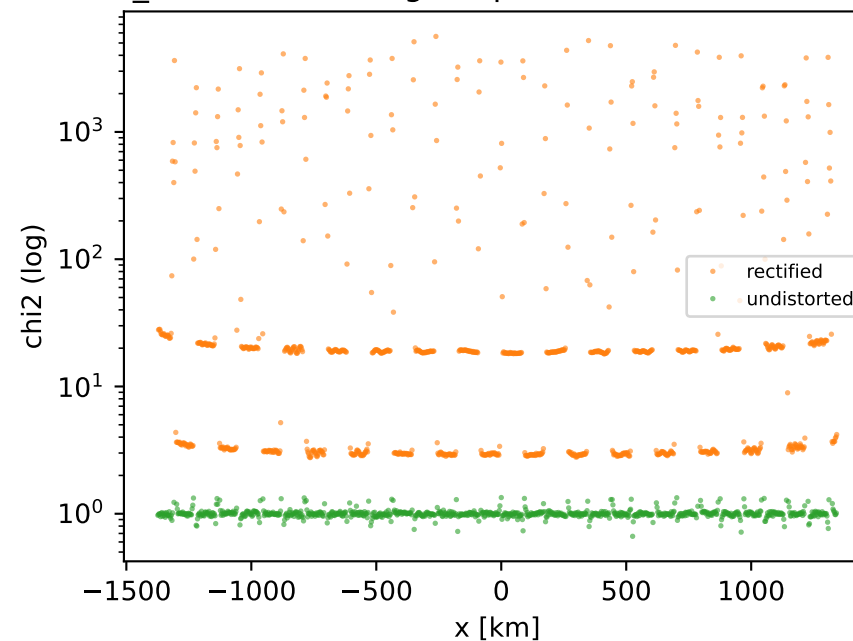
h2o bias vs along-slit position



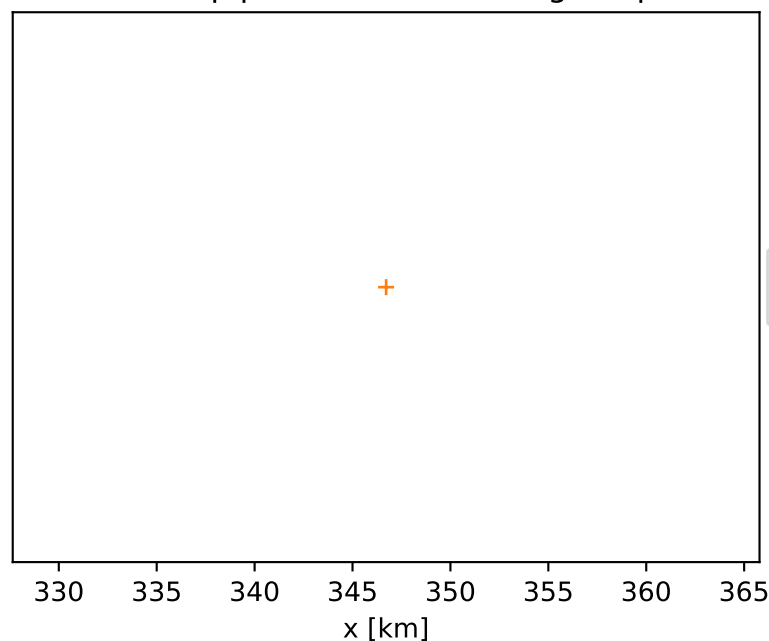
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)



Where each pipeline fails, vs along-slit position



- + rectified stalled (n=1)
- rectified off-detector (n=1)
- undistorted chi2-outlier (n=75)

FPA0+FPA1+FPA2 -- robust bias statistics across scenes/noise (rectified/undistorted)

scene	noise	quantity	pipeline	n	mean	std	median	MAD-sig	n_out
realistic	0	co2	rectified	1017	-13.43	0.947	-13.39	0.7509	0
realistic	0	h2o	rectified	1017	+322.2	597.7	+249.4	826.9	0
realistic	0	p_surface	rectified	1017	-2.657	8.74	-1.121	1.284	180
realistic	0	co2	undistorted	1002	-0.05099	0.2999	-0.07701	0.03372	199
realistic	0	h2o	undistorted	1002	+368.8	628.5	+273.9	896.5	0
realistic	0	p_surface	undistorted	1002	+0.04501	0.06923	+0.06037	0.07135	0
realistic	1	co2	rectified	1018	-13.42	0.9606	-13.4	0.7442	0
realistic	1	h2o	rectified	1018	+320.9	598.9	+250.2	828.8	0
realistic	1	p_surface	rectified	1018	-2.65	8.745	-1.113	1.295	178
realistic	1	co2	undistorted	1002	-0.04554	0.3387	-0.08365	0.2235	0
realistic	1	h2o	undistorted	1002	+368.8	628.4	+274.2	897.6	0
realistic	1	p_surface	undistorted	1002	+0.04891	0.1624	+0.05215	0.1736	0
uniform	0	co2	rectified	1020	-13.5	0.6794	-13.52	0.592	0
uniform	0	h2o	rectified	1020	-51.05	10.76	-51.39	9.251	0
uniform	0	p_surface	rectified	1020	-1.99	1.192	-1.69	0.7805	7
uniform	0	co2	undistorted	1002	-0.0001874	2.802e-06	-0.0001875	3.037e-06	0
uniform	0	h2o	undistorted	1002	-0.000493	3.728e-05	-0.0004938	3.816e-05	0
uniform	0	p_surface	undistorted	1002	-2.897e-05	7.353e-06	-3.071e-05	7.091e-06	0
uniform	1	co2	rectified	1020	-13.49	0.703	-13.52	0.6081	0
uniform	1	h2o	rectified	1020	-50.92	11.4	-51.5	9.665	0
uniform	1	p_surface	rectified	1020	-1.996	1.218	-1.711	0.7937	8
uniform	1	co2	undistorted	1002	+0.005133	0.1657	+0.006247	0.1697	0
uniform	1	h2o	undistorted	1002	+0.04249	2.456	-0.04975	2.467	0
uniform	1	p_surface	undistorted	1002	+0.003875	0.1443	+0.007152	0.151	0
barcode	0	co2	rectified	1021	-13.23	11.55	-13.23	0.8435	91
barcode	0	h2o	rectified	1021	-29.61	147.7	-52.23	10.93	88
barcode	0	p_surface	rectified	1021	-0.8308	8.595	-1.09	2.044	46
barcode	0	co2	undistorted	1002	-0.0001494	4.184e-05	-0.0001811	2.557e-05	0
barcode	0	h2o	undistorted	1002	-0.000323	0.0001895	-0.0004084	0.0002383	0
barcode	0	p_surface	undistorted	1002	-1.292e-06	2.938e-05	-8.522e-06	4.006e-05	0
barcode	1	co2	rectified	1022	-13.32	11.17	-13.27	0.8898	89
barcode	1	h2o	rectified	1022	-28.77	147.3	-52.01	11.97	83
barcode	1	p_surface	rectified	1022	-0.8819	8.581	-1.044	2.013	47
barcode	1	co2	undistorted	927	+0.007615	0.3236	+0.00266	0.2465	0
barcode	1	h2o	undistorted	927	+0.2352	4.739	-0.05955	3.467	0
barcode	1	p_surface	undistorted	927	+0.01622	0.2986	+0.007404	0.2144	0